

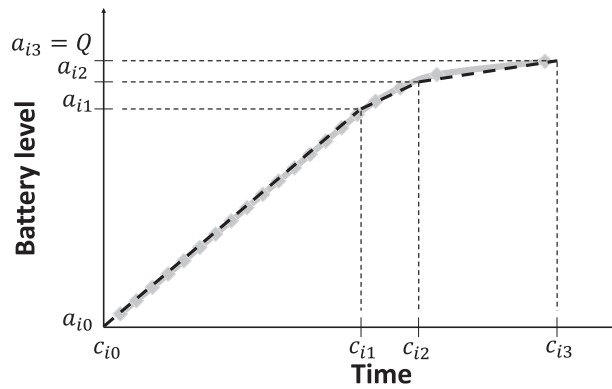


Fig. 3. Real data vs. piecewise linear approximation for a CS of 22 kW charging a battery of 16 kWh.

The remainder of this paper is organized as follows. Section 2 formally introduces the E-VRP-NL. Section 3 describes our hybrid metaheuristic, and Section 4 presents the computational experiments. Finally, Section 5 concludes the paper and discusses future research.

2. Electric vehicle routing problem with nonlinear charging function

2.1. Problem description

Let I be the set of nodes representing the customers, F the set of CSs, and 0 a node representing the depot. Each customer $i \in I$ has a service time p_i . The E-VRP-NL is defined on a directed and complete graph $G = (V, A)$, where $V = \{0\} \cup I \cup F'$ and F' contains the set F and β copies of each CS (i.e., $|F'| = |F| \times (1 + \beta)$). The value of $1 + \beta$ corresponds to the number of times that each CS can be visited. Let $A = \{(i, j) : i, j \in V, i \neq j\}$ be the set of arcs connecting vertices of V . Each arc (i, j) has two associated nonnegative values: a travel time t_{ij} and an energy consumption e_{ij} . The customers are served using an unlimited and homogeneous fleet of EVs. All the EVs have a battery of capacity Q (expressed in kWh) and a maximum tour duration $T_{\max}$. It is assumed that the EVs leave the depot with a fully charged battery, and that all the CSs can handle an unlimited number of EVs simultaneously. Feasible solutions to the E-VRP-NL satisfy the following conditions: each customer is visited exactly once; each route satisfies the maximum-duration limit; each route starts and ends at the depot; and the battery level when an EV arrives at and departs from any vertex is between 0 and Q .

Since the traveled distance is directly related to the energy consumption, most work on E-VRPs with a homogeneous fleet focuses on minimizing the total distance (Schneider et al., 2014; Desaulniers et al., 2016; Hiermann et al., 2016; Keskin and C  tay, 2016). However, this objective function neglects the impact of charging operations. This may lead to solutions that charge the batteries more than needed or charge them even when their level is high. These decisions directly affect the battery's long-term degradation cost (which according to Becker et al., 2009 can be three times the energy cost) and the charging fees at CSs (Bansal, 2015). To better capture the impact of charging operations, in the E-VRP-NL we minimize the total travel and charging time. This objective function was studied by Z  ndorf (2014) and Liao et al. (2016) for related routing problems.

2.2. Modeling of battery charging functions

Each CS $i \in F'$ has a charging mode (e.g., slow, moderate, fast) that is associated with a charging function $g_i(q_i, \Delta_i)$. This function maps the charge level when the vehicle arrives at i (q_i) and the time spent charging at i (Δ_i) to the charge level when the vehicle leaves i . To avoid handling a two-dimensional function, we use the transformation proposed by Z  ndorf (2014). Let $\hat{g}_i(l)$ be the charging function when $q_i = 0$ and the battery is charged for l time units; $g_i(q_i, \Delta_i)$ is estimated as $\hat{g}_i(\Delta_i + \hat{g}_i^{-1}(q_i))$. Note that $\Delta_i = \hat{g}_i^{-1}(o_i) - \hat{g}_i^{-1}(q_i)$, where o_i is the charge level when the vehicle leaves i .

The function $\hat{g}_i(l)$ is concave (Bruglieri et al., 2014; H  imoja et al., 2012; Pelletier et al., 2017) with an asymptote at Q . Similarly to Z  ndorf (2014), we argue that $\hat{g}_i(l)$ can be accurately approximated using piecewise linear functions. We support our claim using the data provided by Uhrig et al. (2015). We fit piecewise linear functions to their data and obtain approximations with an average relative absolute error of 0.90%, 1.24%, and 1.90% for CSs of 11, 22, and 44 kW, respectively. Fig. 3 shows the piecewise linear approximation for a CS i of 22 kW charging a vehicle equipped with a battery of 16 kWh. In the plot, c_{ik} and a_{ik} represent the charging time and the charge level for the breakpoint $k \in B$ of the CS $i \in F'$, where $B = \{0, \dots, b\}$ is the set of breakpoints of the piecewise linear approximation.

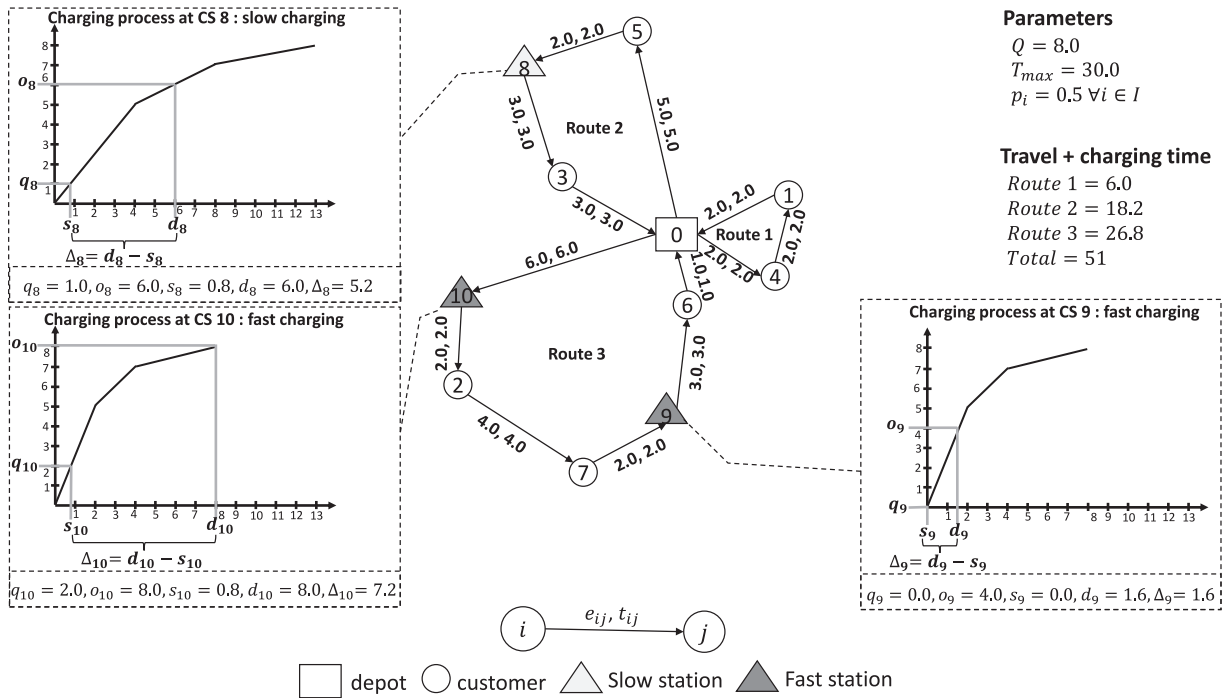


Fig. 4. Example of a feasible E-VRP-NL solution.

2.3. Illustrative example

Fig. 4 presents a numerical example illustrating the E-VRP-NL. The figure depicts a solution to an instance with 7 customers and 3 CSs. The CSs have different technologies (slow and fast), and each technology has a specific piecewise-linear charging function. The charging function maps the battery levels q_i and o_i to the charging times s_i and d_i to estimate the time spent at the CS $i \in F$ (Δ_i). In this example, Route 1 does not visit any CS, because its total energy consumption is less than the battery capacity. Route 2 visits CS 8: the EV arrives at the CS with a battery level $q_8 = 1.0$, and it charges the battery to a level $o_8 = 6.0$. To estimate the time spent at the CS, we use the piecewise-linear charging function: the charging times associated with q_8 and o_8 are $s_i = 0.8$ and $d_i = 6.0$, so the time spent at CS 8 is $\Delta_8 = 6.0 - 0.8 = 5.2$. The duration of Route 2 is the sum of the travel time (13.0), the charging time (5.2), and the service time (1.0), i.e., 19.2, which is less than T_{max} . The cost of this route is 18.2 (travel time + charging time). Finally, Route 3 visits CSs 10 and 9, and it spends $\Delta_{10} = 7.2$ and $\Delta_9 = 1.6$ time units charging in these CSs, respectively.

2.4. Mixed-integer linear programming formulation

To help the reader understand the E-VRP-NL, we now provide a mixed integer linear programming (MILP) formulation of the problem. The MILP uses the following decision variables: variable x_{ij} is equal to 1 if an EV travels from vertex i to j , and 0 otherwise. Variables τ_j and y_j track the time and charge level when the EV departs from vertex $j \in V$. Variables q_i and o_i specify the charge levels when an EV arrives at and departs from CS $i \in F'$, and s_i and d_i are the associated charging times. Variable $\Delta_i = d_i - s_i$ represents the time spent at CS $i \in F'$. Variables z_{ik} and w_{ik} are equal to 1 if the charge level is between $a_{i,k-1}$ and $a_{i,k}$, with $k \in B \setminus \{0\}$, when the EV arrives at and departs from CS $i \in F'$ respectively. Finally, variables α_{ik} and λ_{ik} are the coefficients of the breakpoint $k \in B$ in the piecewise linear approximation, when the EV arrives at and departs from CS $i \in F'$ respectively. The MILP formulation follows:

$$\min \sum_{i,j \in V} t_{ij} x_{ij} + \sum_{i \in F'} \Delta_i \quad (1)$$

subject to

$$\sum_{j \in V, i \neq j} x_{ij} = 1, \quad \forall i \in I \quad (2)$$

$$\sum_{j \in V, i \neq j} x_{ij} \leq 1, \quad \forall i \in F' \quad (3)$$

$$\sum_{j \in V, i \neq j} x_{ji} - \sum_{j \in V, i \neq j} x_{ij} = 0, \quad \forall i \in V \quad (4)$$

$$e_{ij}x_{ij} - (1 - x_{ij})Q \leq y_i - y_j \leq e_{ij}x_{ij} + (1 - x_{ij})Q, \quad \forall i \in V, \forall j \in I, i \neq j \quad (5)$$

$$e_{ij}x_{ij} - (1 - x_{ij})Q \leq y_i - q_j \leq e_{ij}x_{ij} + (1 - x_{ij})Q, \quad \forall i \in V, \forall j \in F', i \neq j \quad (6)$$

$$y_i \geq e_{i0}x_{i0}, \quad \forall i \in V \quad (7)$$

$$y_i = o_i, \quad \forall i \in F' \quad (8)$$

$$y_0 = Q \quad (9)$$

$$q_i \leq o_i, \quad \forall i \in F' \quad (10)$$

$$q_i = \sum_{k \in B} \alpha_{ik} a_{ik}, \quad \forall i \in F' \quad (11)$$

$$s_i = \sum_{k \in B} \alpha_{ik} c_{ik}, \quad \forall i \in F' \quad (12)$$

$$\sum_{k \in B} \alpha_{ik} = \sum_{k \in B \setminus \{0\}} z_{ik}, \quad \forall i \in F' \quad (13)$$

$$\sum_{k \in B \setminus \{0\}} z_{ik} = \sum_{j \in V} x_{ij}, \quad \forall i \in F' \quad (14)$$

$$\alpha_{i0} \leq z_{i1}, \quad \forall i \in F', \quad (15)$$

$$\alpha_{ik} \leq z_{ik} + z_{i,k+1}, \quad \forall i \in F', \forall k \in B \setminus \{0, b\} \quad (16)$$

$$\alpha_{ib} \leq z_{ib}, \quad \forall i \in F' \quad (17)$$

$$o_i = \sum_{k \in B} \lambda_{ik} a_{ik}, \quad \forall i \in F' \quad (18)$$

$$d_i = \sum_{k \in B} \lambda_{ik} c_{ik}, \quad \forall i \in F' \quad (19)$$

$$\sum_{k \in B} \lambda_{ik} = \sum_{k \in B \setminus \{0\}} w_{ik}, \quad \forall i \in F' \quad (20)$$

$$\sum_{k \in B \setminus \{0\}} w_{ik} = \sum_{j \in V} x_{ij}, \quad \forall i \in F' \quad (21)$$

$$\lambda_{i0} \leq w_{i1}, \quad \forall i \in F' \quad (22)$$

$$\lambda_{ik} \leq w_{ik} + w_{i,k+1}, \quad \forall i \in F', \forall k \in B \setminus \{0, b\} \quad (23)$$

$$\lambda_{ib} \leq w_{ib}, \quad \forall i \in F' \quad (24)$$

$$\Delta_i = d_i - s_i, \quad \forall i \in F' \quad (25)$$

$$\tau_i + (t_{ij} + p_j)x_{ij} - T_{\max}(1 - x_{ij}) \leq \tau_j, \quad \forall i \in V, \forall j \in I, i \neq j \quad (26)$$

$$\tau_i + \Delta_j + t_{ij}x_{ij} - (S_{\max} + T_{\max})(1 - x_{ij}) \leq \tau_j, \quad \forall i \in V, \forall j \in F', i \neq j \quad (27)$$

$$\tau_j + t_{j0} \leq T_{\max}, \quad \forall j \in V \quad (28)$$

$$\tau_0 \leq T_{\max} \quad (29)$$

$$x_{ij} = 0, \quad \forall i, j \in F' : m_{ij} = 1 \quad (30)$$

$$\tau_i \geq \tau_j, \quad \forall i, j \in F' : m_{ij} = 1, j \leq i \quad (31)$$

$$\tau_j \leq T_{\max} \sum_{i \in V} x_{ij}, \quad \forall j \in F' \quad (32)$$

$$\sum_{i \in V} x_{ih} \geq \sum_{j \in V} x_{jf}, \quad \forall h, f \in F' : m_{hf} = 1, h \leq f \quad (33)$$

$$x_{ij} \in \{0, 1\}, \quad \forall i, j \in V, i \neq j \quad (34)$$

$$\tau_i \geq 0, y_i \geq 0 \quad \forall i \in V \quad (35)$$

$$z_{ik} \in \{0, 1\}, w_{ik} \in \{0, 1\}, \quad \forall i \in F', \forall k \in B \setminus \{0\} \quad (36)$$

$$\alpha_{ik} \geq 0, \lambda_{ik} \geq 0, \quad \forall i \in F', \forall k \in B \quad (37)$$

$$q_i \geq 0, o_i \geq 0, s_i \geq 0, d_i \geq 0, \Delta_i \geq 0, \quad \forall i \in F' \quad (38)$$

The objective function (1) seeks to minimize the total time (travel times plus charging times). Constraints (2) ensure that each customer is visited once. Constraints (3) ensure that each CS copy is visited at most once. Constraints (4) impose the flow conservation. Constraints (5) and (6) track the battery charge level at each vertex. Constraints (7) ensure that if the EV travels between a vertex and the depot, it has sufficient energy to reach its destination. Constraints (8) reset the battery tracking to o_i upon departure from CS $i \in F'$. Constraint (9) ensures that the battery charge level is Q at the depot. Constraints (10) couple the charge levels when an EV arrives at and departs from any CS. Constraints (11)–(17) define the charge level (and its corresponding charging time) when an EV arrives at CS $i \in F'$ (based on the piecewise linear approximation of the charging function). Similarly, constraints (18)–(24) define the charge level (and its corresponding charging time) when an EV departs from CS $i \in F'$. Constraints (25) define the time spent at any CS. Constraints (26) and (27) track the departure time at each vertex, where $S_{\max} = \max_{i \in F'} \{c_{ib}\}$. Constraints (28) and (29) ensure that the EVs return to the depot no later than $T_{\max}$. Constraints (30)–(33) help to avoid the symmetry generated by the copies of the CSs. Parameter m_{ij} is equal to 1 if i and $j \in F'$ represent the same CS. Finally, constraints (34)–(38) define the domain of the decision variables.

3. Solving the E-VRP-NL

Lenstra and Rinnooy Kan (1981) demonstrated that the classical VRP, commonly known as the CVRP, is NP-hard. Since the CVRP is a special case of our E-VRP-NL, the latter is also NP-hard. We therefore propose a metaheuristic approach. Like many metaheuristics for other VRPs, our approach explores new solutions by building new routes or applying changes (moves) to existing ones. In this process, the algorithm makes sequencing and charging decisions. The former fix the order in which the route visits its assigned customers, while the latter determine where and how much to charge the EV serving the route. Sequencing and charging decisions can be made either simultaneously or in two phases (sequencing first, charging second). We use the latter option. To make charging decisions, we solve what we call the *fixed route vehicle charging problem*, or simply FRVCP. In a nutshell, the problem consists in (i) inserting charging stations into a fixed sequence of customers and (ii) deciding how much to charge at each inserted station. Since the FRVCP plays a key role in our approach, we discuss it in the next subsection before introducing our metaheuristic.